

FROSTMASTER

Pro Series 600

OWNER'S MANUAL

Model FM-600XL | French Door Refrigerator

Total Capacity: 28.5 Cu. Ft. | Energy Star Certified

Please read and retain this manual for future reference.

Document Version 2.1 | Published January 2025

Table of Contents

1.	Safety Information & Warnings	3
2.	Product Overview & Specifications	4
3.	Installation & Setup	5
4.	Getting to Know Your Refrigerator	7
5.	Temperature & Control Settings	9
6.	Food Storage Guidelines	11
7.	Ice Maker & Water Dispenser	13
8.	Energy Saving Tips	14
9.	Cleaning & Maintenance	15
10.	Troubleshooting Guide	17
11.	Error Codes	18
12.	Warranty & Customer Support	19

1. Safety Information & Warnings

Read all instructions carefully before operating your refrigerator. Failure to follow these safety guidelines may result in property damage, personal injury, or death.

⚠ DANGER – Read Before Use

NEVER operate this appliance with a damaged cord or plug.
NEVER store flammable materials, gasoline, or explosive substances near the refrigerator.
NEVER allow children to play inside or climb on the refrigerator.
NEVER attempt to repair internal components yourself — contact a certified technician.
NEVER place the refrigerator near direct heat sources such as ovens or direct sunlight.

⚡ ELECTRICAL SAFETY

This appliance requires a grounded 115V AC, 60 Hz electrical outlet.
Use a dedicated circuit — do not share with other high-wattage appliances.
Ensure the power cord is never pinched behind or under the unit.
If the power supply cord is damaged, it must be replaced by qualified personnel.
Unplug the unit before cleaning or performing any maintenance.

Child Safety

Remove all doors, shelves, and drawers before discarding the unit. This prevents children from becoming trapped inside, which can cause suffocation. If keeping the unit temporarily, ensure it is stored where children cannot access it.

Refrigerant Safety

This refrigerator uses R-600a (isobutane) refrigerant. While environmentally friendly with low global warming potential, it is mildly flammable. Do not puncture refrigerant tubing or expose to open flame. In the event of a refrigerant leak, ventilate the area immediately and contact a certified technician.

2. Product Overview & Specifications

The FrostMaster Pro Series 600 (Model FM-600XL) is a premium French door refrigerator designed for modern kitchens. It combines advanced multi-zone cooling technology, smart connectivity, and a sleek stainless steel finish.

Technical Specifications

Specification	Value
Model Number	FM-600XL
Configuration	French Door with Bottom Freezer Drawer
Total Capacity	28.5 Cu. Ft. (19.8 Refrigerator + 8.7 Freezer)
Dimensions (H × W × D)	69.75" × 35.75" × 34.125" (with handles)
Net Weight	335 lbs (152 kg)
Power Supply	115V AC, 60 Hz, 7.0 A
Annual Energy Consumption	630 kWh/year
Refrigerant Type	R-600a (Isobutane), 1.8 oz
Noise Level	≤ 40 dB
Compressor Type	Inverter Linear Compressor
Smart Features	Wi-Fi Connected, App Control, Voice Assistant Compatible
Door Alarm	Yes (audible + visual)
Water/Ice Dispenser	Dual — Crushed & Cubed Ice, Filtered Water
Color / Finish	Premium Stainless Steel, Fingerprint Resistant

Included Accessories

- 2× adjustable glass shelves
- 1× deli drawer
- 2× full-width humidity-controlled crisper drawers
- 3× door bins (2 adjustable, 1 gallon-sized)
- 1× freezer basket
- Water filter (initial installation)
- Ice tray / ice maker components
- Power cord (6 ft)
- Installation and leveling hardware kit

- This owner's manual

3. Installation & Setup

3.1 Before You Begin

Before installing your refrigerator, inspect the unit for any shipping damage. If damage is present, document it and contact FrostMaster Customer Support immediately. Do not install a damaged appliance.

IMPORTANT: Allow 24 hours

After moving the refrigerator to its installation location, allow it to stand upright for at least 24 hours before plugging it in. This allows the refrigerant and compressor oil to settle, extending the life of the compressor.

3.2 Location Requirements

- Select a level, firm surface capable of supporting the refrigerator's weight.
- Ensure ambient temperature is between 55°F (13°C) and 110°F (43°C).
- Allow minimum clearances: 1 inch on each side, 1 inch at the top, and 2 inches at the back.
- Keep away from direct sunlight, ovens, dishwashers, and heat registers.
- Ensure access to a grounded 115V electrical outlet within 6 feet.
- Water supply line (1/4 inch) must be accessible for ice maker and water dispenser.

3.3 Installation Steps

1. Remove all packaging materials, foam inserts, and tape from inside and outside the unit.
2. With assistance, move the refrigerator to its final location (do not tilt more than 45°).
3. Level the refrigerator using the adjustable front rollers. Use a bubble level on top.
4. Connect the water supply line to the inlet valve at the rear bottom of the unit.
5. Plug the power cord into the grounded outlet. Do not use an extension cord.
6. Press the POWER button and allow 3–4 hours for the unit to reach operating temperature.
7. Install the water filter (see Section 7) before using water dispenser or ice maker.

3.4 Leveling the Refrigerator

A properly leveled refrigerator ensures doors seal correctly, prevents water pooling, and reduces compressor wear. Use a standard bubble level placed on top of the unit.

Adjustment	Method
Tilt front-to-back	Turn front leveling rollers: clockwise to raise, counter-clockwise to lower
Tilt left-to-right	Adjust left or right roller independently
Door alignment	Adjust hinge pin on lower door hinge using provided hex wrench
Anti-tip bracket	Secure the provided anti-tip bracket to wall stud behind the unit

4. Getting to Know Your Refrigerator

The FM-600XL features multiple compartments and zones, each optimized for specific food types. Understanding the layout helps you store food at the correct temperature and humidity.

4.1 Exterior Features

Component	Description
Control Panel	LED touchscreen located on the left refrigerator door; controls all settings
Water & Ice Dispenser	Located on the left door exterior; dispenses filtered water, cubed & crushed ice
Door Handles	Premium stainless steel recessed handles on all 3 doors
Fingerprint-Resistant Finish	Nano-coating prevents smudges; wipe with damp microfiber cloth
Door Alarm Indicator	Red LED flashes if door is open more than 2 minutes
USB Charging Port	1× USB-A port on right side panel (5V / 1A output)

4.2 Refrigerator Compartment

Feature	Function
Upper Shelf (Fixed)	Ideal for beverages, tall items, leftovers in containers
Middle Shelf (Adjustable)	Adjusts in 1-inch increments; suited for dairy and prepared foods
Lower Shelf (Adjustable)	Close to crisper drawers; use for foods that benefit from cooler temps
Deli Drawer	Narrow pull-out drawer with dedicated temperature zone; for meats & cheeses
Crisper Drawer A (High Humidity)	Maintains ~90% RH; optimal for leafy greens and herbs
Crisper Drawer B (Low Humidity)	Maintains ~50% RH; optimal for fruits and root vegetables
Door Bins	Store condiments, juices, and small bottles; not suitable for raw meat
Gallon Door Bin	Accommodates 1-gallon jugs, 2-liter bottles

4.3 Freezer Compartment

Feature	Function
Upper Freezer Shelf	Slim wire shelf for flat frozen items like pizzas and trays
Lower Freezer Basket	Deep pull-out basket for bulk frozen goods, meats, and ice cream
Ice Maker Bin	Located on the right side; remove to access ice maker components
Express Freeze Zone	Dedicated zone that drops to -4°F (-20°C) for rapid freezing

4.4 Control Panel Overview

The LED touchscreen control panel provides access to all settings. The panel locks automatically after 30 seconds of inactivity to prevent accidental changes.

Button / Icon	Function
FRIDGE TEMP +/-	Adjusts refrigerator temperature in 1°F increments
FREEZER TEMP +/-	Adjusts freezer temperature in 1°F increments
ICE TYPE	Toggles between Off / Cubed / Crushed
WATER FILTER	Shows filter life; hold 3 seconds to reset after replacement
DOOR ALARM	Toggles door-open alarm on/off
SMART COOL	Activates AI-driven temperature optimization mode
VACATION MODE	Raises temp to save energy when away for extended periods
LOCK	Panel child lock; hold 3 seconds to activate/deactivate
Wi-Fi	Connects to FrostMaster app; indicator light shows connection status

5. Temperature & Control Settings

5.1 Recommended Temperature Settings

Maintaining proper temperatures is critical for food safety and preserving quality. The FDA recommends the following refrigerator and freezer temperatures:

Compartment	Recommended Temperature
Refrigerator (Main)	35°F – 38°F (1.7°C – 3.3°C)
Refrigerator (Deli Drawer)	28°F – 32°F (-2.2°C – 0°C)
Freezer	0°F (-18°C)
Express Freeze Zone	-4°F (-20°C) — for initial freezing only

NOTE: Temperature Stabilization

After changing temperature settings, allow 24 hours for the refrigerator to stabilize at the new temperature. Do not adjust settings again until 24 hours have passed.

5.2 Smart Cool Technology

Smart Cool is an AI-powered feature that learns your household's usage patterns over 2–4 weeks and automatically adjusts cooling to maintain optimal temperature stability while minimizing energy use. Smart Cool accounts for:

- Door opening frequency and duration
- Ambient kitchen temperature and humidity
- Amount of food stored in each zone
- Time of day and usage patterns

To enable Smart Cool: press the SMART COOL button on the control panel. The indicator light will blink during the learning period and turn solid when fully active.

5.3 Vacation Mode

Vacation Mode is designed for trips lasting 2 weeks or longer. It raises the refrigerator temperature to 50°F (10°C) and reduces compressor cycling to save energy. The freezer remains at its normal setting.

IMPORTANT: Before Enabling Vacation Mode

Remove all perishable foods from the refrigerator section.

Leave some food items (sealed, non-perishable) in the refrigerator to help maintain thermal mass.

Ensure the water supply valve is open if ice maker will remain active.

Do NOT use Vacation Mode if fresh food remains in the refrigerator.

5.4 Express Freeze

Express Freeze rapidly cools the entire freezer compartment to -4°F (-20°C) for up to 24 hours. This is ideal for freezing large quantities of fresh food quickly, preserving texture and nutritional value.

To activate: Press and hold the FREEZER TEMP button for 3 seconds until the snowflake icon appears. Express Freeze deactivates automatically after 24 hours and returns to the set temperature.

6. Food Storage Guidelines

Proper food storage prevents spoilage, reduces food waste, and keeps your family safe. Follow these guidelines for each area of the refrigerator.

6.1 Refrigerator Shelves — Storage Tips

- Store ready-to-eat foods (leftovers, cooked meats) on upper shelves.
- Keep raw meats in sealed containers on the lowest shelf to prevent drip contamination.
- Do not overcrowd shelves — cold air must circulate for even cooling.
- Allow hot foods to cool to room temperature before placing in the refrigerator.
- Always cover or seal open containers to prevent odor transfer.

6.2 Crisper Drawers

Drawer Setting	Best For
High Humidity (90% RH)	Asparagus, broccoli, leafy greens, herbs, cucumbers
Low Humidity (50% RH)	Apples, pears, grapes, citrus, stone fruits, avocados
Avoid Mixing	Ethylene-producing fruits (apples, pears) accelerate ripening of vegetables

6.3 Food Storage Duration Reference

Food Item	Recommended Storage Time
Raw Poultry (chicken, turkey)	1–2 days in refrigerator; 9–12 months frozen
Raw Ground Beef / Pork	1–2 days in refrigerator; 3–4 months frozen
Raw Steaks / Roasts	3–5 days in refrigerator; 4–12 months frozen
Cooked Leftovers	3–4 days in refrigerator; 2–3 months frozen
Fresh Fish / Seafood	1–2 days in refrigerator; 2–3 months frozen
Eggs (in shell)	3–5 weeks in refrigerator
Hard Cheese	3–4 weeks in refrigerator
Soft Cheese	1–2 weeks in refrigerator
Milk / Cream	1 week after opening
Fresh Vegetables (most)	5–7 days in crisper

Fruits (most)	3–5 days
Opened Deli Meats	3–5 days
Opened Condiments	1–3 months (varies by product)

6.4 What NOT to Refrigerate

Foods That Should NOT Be Refrigerated

Whole tomatoes (cold damages texture and flavor — store at room temp)
Whole uncut onions (moisture causes molding — store in dry, cool pantry)
Garlic (refrigeration can cause sprouting or softening)
Potatoes (cold converts starch to sugar — store in dark, cool pantry)
Honey (crystallizes in cold temperatures — room temperature is ideal)
Bananas (cold turns skin black — refrigerate only once fully ripe)
Bread (refrigerator air dries out bread faster — use bread box or freeze)

7. Ice Maker & Water Dispenser

7.1 Water Filter Installation & Replacement

The FM-600XL uses a push-in water filter (Model: FM-WF3) located in the upper right interior of the refrigerator compartment. The filter should be replaced every 6 months or after 300 gallons of use, whichever comes first.

8. Open the filter housing by pressing the release tab.
9. Remove the old filter by pulling straight out.
10. Unwrap the new filter and insert it firmly until it clicks.
11. Run 2 gallons of water through the dispenser to flush the new filter.
12. Reset the filter indicator by holding the WATER FILTER button for 3 seconds.

Filter Status Indicators

Green light: Filter is functioning normally

Yellow light: Filter is approaching end of life (replace within 30 days)

Red light: Filter must be replaced immediately — water quality may be compromised

7.2 Ice Maker Operation

The automatic ice maker is located in the upper section of the freezer compartment. Once the freezer reaches 15°F (-9.4°C), the ice maker will begin producing ice automatically.

Specification	Detail
Daily Ice Production	Approximately 8–10 lbs per day
Ice Bin Capacity	4.4 lbs
Ice Types	Cubed and Crushed (selectable via control panel)
Time to First Ice	12–24 hours after installation
Minimum Freezer Temp	15°F (-9.4°C) for ice production to begin

7.3 Ice Maker Troubleshooting

Issue	Possible Cause / Solution
No ice production	Check water supply valve is open; ensure freezer is at 0°F; check filter

Small or hollow ice cubes	Low water pressure — minimum 20 PSI required; replace filter
Ice has bad taste/odor	Replace water filter; discard old ice and allow new ice to cycle
Ice maker makes noise	Normal — popping/cracking sounds occur during ice harvest cycle
Ice clumps together	Dispense ice daily to prevent clumping; check door seal

8. Energy Saving Tips

The FM-600XL is Energy Star certified, consuming approximately 630 kWh per year — well below the federal standard. You can reduce energy usage further by following these best practices:

8.1 Placement & Usage

- Keep the refrigerator away from heat sources — every degree of ambient heat increases energy use by ~2.5%.
- Ensure proper ventilation clearances are maintained (see Section 3).
- Minimize door openings — plan what you need before opening.
- Let hot foods cool to room temperature before placing in the fridge.
- Keep the freezer at least 75% full — a fuller freezer retains cold better.
- Use vacuum-sealed containers to minimize moisture transfer.

8.2 Settings & Features

- Enable Smart Cool mode — can reduce energy use by up to 12% over manual settings.
- Use Vacation Mode during extended absences.
- Set refrigerator no colder than 35°F (1.7°C) — colder settings waste energy without benefit.
- Turn off the ice maker if you rarely use ice — it uses approximately 18–20 kWh per year.
- Clean condenser coils twice a year (see Section 9) to maintain efficiency.

Energy Savings Estimate

Following all energy-saving tips above can reduce your annual energy consumption by 10–15%, saving approximately \$10–\$18 per year at average US electricity rates.

Over 10 years, this equals \$100–\$180 in savings while reducing CO2 emissions.

9. Cleaning & Maintenance

9.1 Interior Cleaning

Before Cleaning

Always unplug the refrigerator before thorough interior cleaning.
Never use abrasive cleaners, bleach, or scouring pads on interior surfaces.
Do not allow water to pool in the bottom of the refrigerator.

13. Remove all food items and store them in a cooler.
14. Remove all shelves, drawers, and bins — wash separately in warm soapy water.
15. Wipe interior walls with a solution of 1 tablespoon baking soda per quart of warm water.
16. Dry all surfaces thoroughly before replacing components.
17. Clean door gaskets with warm, soapy water and a soft cloth.
18. Replace all components and food items once interior is dry.

9.2 Exterior Cleaning

Wipe the stainless steel exterior with a damp microfiber cloth, following the grain of the steel. For stubborn marks, use a small amount of stainless steel cleaner. Avoid chlorine-based cleaners.

Clean the control panel with a lightly dampened cloth — never spray cleaner directly onto the panel. Activate Panel Lock before cleaning to prevent accidental input.

9.3 Condenser Coil Cleaning

Condenser coils are located behind the lower kick plate at the front bottom of the unit. Dust buildup on coils forces the compressor to work harder, increasing energy use and reducing lifespan. Clean every 6 months or more frequently if you have pets.

19. Unplug the refrigerator.
20. Remove the lower kick plate (snap off from the bottom).
21. Use a coil brush or vacuum attachment to remove dust from the coils.
22. Replace the kick plate and plug the unit back in.

9.4 Maintenance Schedule

Task	Frequency
Wipe interior surfaces	Monthly
Clean door gaskets	Monthly
Replace water filter	Every 6 months (or 300 gallons)
Clean condenser coils	Every 6 months (more often with pets)
Inspect door seals for wear/cracks	Every 6 months
Clean ice maker bin	Every 3 months
Defrost freezer (if ice buildup > 0.25 inch)	As needed
Full deep-clean (interior + exterior)	Annually
Professional service check	Every 2–3 years

10. Troubleshooting Guide

Before contacting customer support, consult this troubleshooting guide. Most common issues can be resolved without a service call.

Problem	Possible Causes & Solutions
Refrigerator not cooling	Check power connection. Ensure condenser coils are clean. Verify temperature settings. Allow 24 hrs after initial startup.
Refrigerator too cold / freezing food	Raise refrigerator temp by 2°F. Move sensitive items away from air vents. Check Smart Cool settings.
Excessive frost buildup in freezer	Check door seal integrity. Ensure freezer door is fully closing. Manually defrost and inspect.
Unusual noises: Gurgling/bubbling	Normal — refrigerant circulating through coils. No action needed.
Unusual noises: Clicking/popping	Normal — thermal expansion of components during temperature changes.
Unusual noises: Loud humming/vibrating	Ensure refrigerator is level. Check items on top of unit. Ensure back is not touching wall.
Water on floor / leaking	Check water supply line connections. Inspect drain pan (under unit). Clear clogged defrost drain.
Ice maker not producing ice	Verify water supply valve is open. Check water filter. Ensure freezer temp is at 0°F.
Bad odors	Check for expired food. Clean interior with baking soda solution. Place open box of baking soda inside.
Door not sealing properly	Clean door gasket. Check for food items blocking door. Adjust leveling rollers.
Control panel unresponsive	Check if Child Lock is activated (hold LOCK button 3 secs). Power cycle the unit.
Wi-Fi connectivity issues	Ensure 2.4 GHz Wi-Fi is available. Re-pair via FrostMaster app. Check router distance.
Condensation on exterior	Normal in high humidity environments. Increase indoor ventilation.
Lights not working	Check if door switch is stuck. Replace LED light module (consult service center).

11. Error Codes

When an error occurs, the control panel displays a code. Use this reference to identify and resolve the issue. If the problem persists after following the recommended action, contact FrostMaster Support.

Error Code	Meaning & Recommended Action
E1	Refrigerator Sensor Error — Unplug for 10 minutes and restart. If persists, contact service.
E2	Freezer Sensor Error — Same as E1.
E3	Defrost Sensor Error — Allow defrost cycle to complete. If alarm continues, contact service.
E4	Evaporator Fan Error — Check for ice obstruction in freezer fan area. Defrost manually if needed.
E5	Condenser Fan Error — Check coils are clean and fan spins freely. Contact service if persistent.
E6	Ice Maker Sensor Error — Remove ice bin, check for obstructions. Reset ice maker.
E7	Water Valve Error — Check water supply line pressure (min 20 PSI). Replace valve if needed.
E8	Door Sensor Error — Clean door hinge contacts. Check door alignment.
E9	Control Board Communication Error — Power cycle unit. Contact service if recurrent.
dF	Forced Defrost Mode Active — Normal operation. Wait for cycle to complete (~30 min).
OF	Refrigerator Off Mode — Check if unit was manually powered off. Press POWER to restart.
HI	Temperature Too High — Door may have been open. Wait for temperature to stabilize.
LO	Temperature Too Low — Adjust temperature settings. Check for sensor malfunction.

How to Clear an Error Code

1. Address the root cause of the error (see table above).
2. Hold the POWER button for 5 seconds to perform a soft reset.
3. If the error code returns within 1 hour, contact FrostMaster Support for a service appointment.
4. Always note the error code before resetting — the technician will need this information.

12. Warranty & Customer Support

12.1 Limited Warranty Coverage

FrostMaster Appliances warrants the FM-600XL against defects in materials and workmanship under normal household use and conditions. This warranty is valid only for the original purchaser and is non-transferable.

Component	Warranty Period
Parts & Labor (general)	1 Year from date of purchase
Sealed Refrigeration System	5 Years (compressor, evaporator, condenser, tubing)
Inverter Linear Compressor	10 Years (parts only after year 5)
Cosmetic / Appearance Issues	30 Days (dents, scratches, discoloration)
Water Filter	Not covered (consumable)
Light Bulbs / LED Strips	90 Days

12.2 What Is Not Covered

- Damage caused by accident, misuse, abuse, or acts of nature
- Service by unauthorized technicians or use of non-genuine parts
- Installation not in accordance with this manual
- Cosmetic damage (scratches, chips, dents) occurring after delivery
- Normal wear and tear, including door gaskets, drawers, and shelves
- Refrigerant recharge if not caused by a manufacturing defect
- Damage from power surges or incorrect electrical supply
- Food loss due to any mechanical or electrical failure

12.3 How to Obtain Warranty Service

23. Keep your proof of purchase (receipt or invoice) in a safe place.
24. Note the model number (FM-600XL) and serial number (found on label inside left door frame).
25. Contact FrostMaster Customer Support using the information below.
26. A service representative will schedule an in-home appointment.
27. Authorized service technicians will be dispatched — do not attempt repairs yourself.

FrostMaster Customer Support

Phone (US & Canada): 1-800-555-FROST (1-800-555-3767)

Monday – Friday, 8:00 AM – 8:00 PM EST

Saturday, 9:00 AM – 5:00 PM EST

Email: support@frostmaster.com

Online Support Portal: www.frostmaster.com/support

App: FrostMaster App (iOS & Android)

Parts & Accessories: www.frostmaster.com/parts

12.4 Product Registration

Register your FM-600XL online at www.frostmaster.com/register within 30 days of purchase to activate your full warranty, receive product updates, and get early access to new features. You will need your model number, serial number, and date of purchase.

Thank you for choosing FrostMaster. We are committed to delivering lasting quality, energy efficiency, and innovative technology to your home. Should you have any questions not addressed in this manual, our support team is ready to assist.

FrostMaster Pro Series 600 | Model FM-600XL | Owner's Manual | Version 2.1
© 2025 FrostMaster Appliances, Inc. All rights reserved.